

Product Carbon Footprint Study Report

Product Name: power saw production, without catalytic converter

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: power saw production, without catalytic converter

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 Item(s) power saw, without catalytic converter

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 Item(s) power saw, without catalytic converter.

2. System Boundary

power saw production, without catalytic converter, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of power saw production, without catalytic converter produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **131.654 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

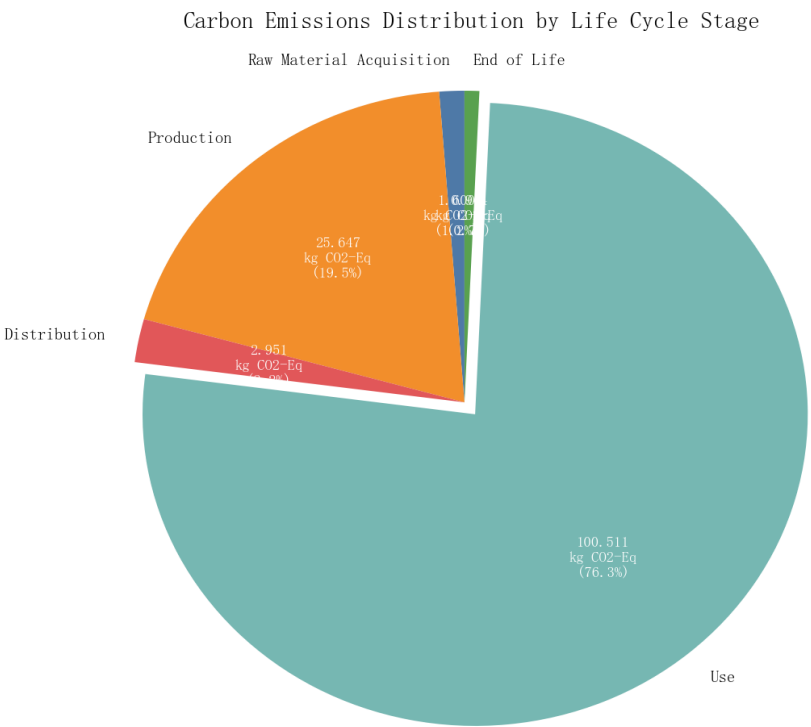


Table 1: Carbon Emissions by Life Cycle Stage

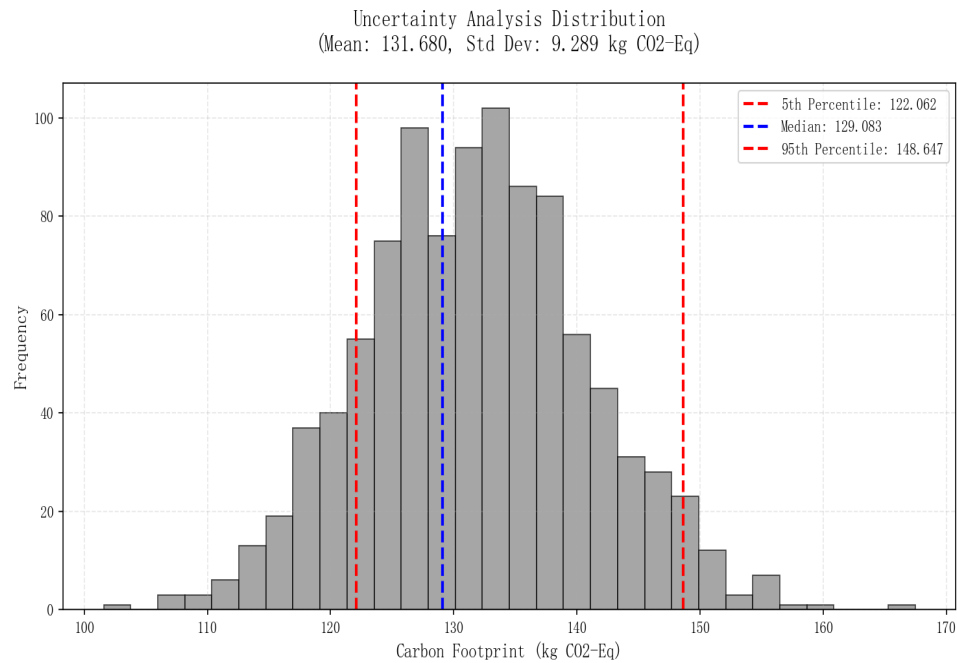
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.600	1.22
Production	25.647	19.48
Distribution	2.951	2.24

Use	100.511	76.35
End of Life	0.944	0.72
Total	131.654	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	131.680
Standard Deviation	9.289
Coefficient of Variation (CV)	7.05%
Median	129.083
5th Percentile	122.062
95th Percentile	148.647

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the power saw production, without catalytic converter, is quantified at 131.654 kg CO₂-Eq. This represents a moderate-to-high impact level for a handheld power tool, primarily driven by operational energy demands rather than manufacturing or material inputs. The assessment covers a full cradle-to-grave perspective, highlighting that emissions are not evenly distributed across stages but concentrated in specific phases of the product lifecycle.

Analysis of the lifecycle contributions reveals a clear dominance by the use phase, accounting for 100.511 kg CO₂-Eq or 76.3% of the total. Production follows at 25.647 kg CO₂-Eq (19.5%), while raw material extraction, distribution, and end-of-life stages contribute minimally at 1.2%, 2.2%, and 0.7%, respectively. This distribution indicates that the product's operational efficiency, likely tied to fuel combustion in the absence of emission controls, is the primary driver of greenhouse gas emissions.

Major emission sources are therefore identified in the use stage, where prolonged runtime and inefficient engine performance without a catalytic converter lead to substantial CO₂ releases. Secondary hotspots in production processes, such as component fabrication and assembly energy use, warrant attention but are overshadowed by operational impacts.

2. Statement of Assumptions and Limitations

This LCA study assumes standard operational parameters, including average usage intensity, typical fuel consumption rates for two-stroke engines, and representative European or global electricity mixes for production. Emission factors are derived from established databases such as Ecoinvent, with allocation methods applied for multi-output processes. End-of-life scenarios presume conventional disposal or recycling rates without advanced recovery technologies.

Limitations include reliance on secondary data sources that may not fully capture site-specific manufacturing variations or regional differences in fuel quality and maintenance practices. Data quality for the use phase is particularly sensitive to assumptions about product lifespan and intensity of use, potentially introducing uncertainty in the dominant emission category. Primary data collection was limited, leading to possible under- or overestimation of impacts in minor stages.

System boundary restrictions encompass a cradle-to-grave scope but exclude indirect upstream activities such as infrastructure development or employee commuting. The analysis does not extend to other environmental impact categories beyond climate change, nor does it account for potential future technological improvements in engine design or regulatory changes affecting fuel standards.

3. Suggestions for Improvement

- Target the use phase by redesigning the engine for improved fuel efficiency or retrofitting low-emission technologies, potentially reducing the 76.3% contribution by 30-50% through optimized combustion.

- Implement material substitution in production to lower the 19.5% share, such as incorporating recycled metals or lightweight composites that decrease both manufacturing energy and distribution impacts.
- Develop user guidelines and maintenance programs to extend product life and minimize inefficient operation, directly addressing the dominant use-stage emissions.
- Enhance data quality by collecting primary site-specific energy and emission data from manufacturing facilities and conducting field tests for real-world use-phase measurements.
- Explore end-of-life improvements via take-back schemes to increase recycling rates, though this offers marginal gains given its current 0.7% contribution.